

Lumped-Parameter Modeling and Calibration of Hexapole Electromagnetic Actuators for Magnetic Flux Generation:

Hexapole Electromagnetic Actuators: Lumped-Parameter Modeling and Flux Generation Calibration:

Magnetic Flux Generation via Hexapole Electromagnetic Actuators: Lumped-Parameter Modeling and Calibration Approach:

1. Data acquisition:

- Select N_p points $\mathbf{p}_i$ (3×1) for $i = 1, 2, \dots, N_p$ in the work space.
- Perform N_I Finite Element Method (FEM) simulations using the input current vectors $\mathbf{I}_j = [I_{j1} \ I_{j2} \ I_{j3} \ I_{j4} \ I_{j5} \ I_{j6}]^T$ for $j = 1, 2, \dots, N_I$, ensuring that the combined current matrix $\mathbf{F} = [\mathbf{I}_1 \ \mathbf{I}_2 \ \dots \ \mathbf{I}_{N_I}]$ has a rank of six. For each simulation.
- Record the magnetic flux density $\mathbf{b}_{ij} = \mathbf{b}^{FEM}(\mathbf{p}_i, \mathbf{I}_j)$ (3×1) for $i = 1, 2, \dots, N_p$.
- Record the sensor voltage v_{jk} for $k = 1, 2, \dots, 6$ and for $j = 1, 2, \dots, N_I$.

2. Modeling hexapole charge location:

- Actuator coordinate system: $[\hat{\mathbf{u}} \ \hat{\mathbf{v}} \ \hat{\mathbf{w}}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
- 18-parameter model: Introduce a characteristic length scale ℓ and a 1×17 bias vector $\mathbf{e}^T$, relating the geometry of the hexapole tweezers to the discrete locations of the six magnetic charges in the lumped-parameter model.

$$\mathbf{e}^T = \left[\underbrace{e_{1u} \ e_{1v} \ e_{1w}}_{\mathbf{e}_1^T} \ \underbrace{e_{2u} \ e_{2v} \ e_{2w}}_{\mathbf{e}_2^T} \ \dots \ \underbrace{e_{5u} \ e_{5v} \ e_{5w}}_{\mathbf{e}_5^T} \ e_{6u} \ e_{6v} \right]$$

$$e_{6w} = e_{1u} - e_{2u} + e_{3v} - e_{4v} + e_{5w}$$

c. Normalized charge location:

$$\bar{\mathbf{p}}_c(\mathbf{e}) = [\hat{\mathbf{u}} + \mathbf{e}_1 \quad -\hat{\mathbf{u}} + \mathbf{e}_2 \quad \hat{\mathbf{v}} + \mathbf{e}_3 \quad -\hat{\mathbf{v}} + \mathbf{e}_4 \quad \hat{\mathbf{w}} + \mathbf{e}_5 \quad -\hat{\mathbf{w}} + \mathbf{e}_6]$$

$$= \begin{bmatrix} 1 + e_{1u} & -1 + e_{2u} & e_{3u} & e_{4u} & e_{5u} & e_{6u} \\ e_{1v} & e_{2v} & 1 + e_{3v} & -1 + e_{4v} & e_{5v} & e_{6v} \\ e_{1w} & e_{2w} & e_{3w} & e_{4w} & 1 + e_{5w} & -1 + (e_{6w}) \end{bmatrix}$$

d. Charge location: $\mathbf{p}_c = \ell \cdot \bar{\mathbf{p}}_c$

3. Position-dependent 3×6 dimensionless flux generation matrix $\mathbf{S}(\mathbf{p}/\ell)$:

$$\bar{\mathbf{p}}_i = \mathbf{p}_i / \ell \text{ for } i = 1, 2, \dots, N_p.$$

$$\mathbf{s}(\mathbf{p}_i / \ell, \hat{\mathbf{e}}) = \left[\underbrace{\frac{\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_{c1}}{\|\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_{c1}\|^3}}_{s_{i1}} \quad \underbrace{\frac{\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_{c2}}{\|\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_{c2}\|^3}}_{s_{i2}} \quad \dots \quad \underbrace{\frac{\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_{c6}}{\|\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_{c6}\|^3}}_{s_{i6}} \right] (3 \times 6) \text{ for } i = 1, 2, \dots, N_p.$$

4. Cost function:

$$\mathbf{b}_{ij} \approx \mathbf{S}(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) \left(\frac{{}^B \hat{\mathbf{g}}_I \bar{\mathbf{K}} \mathbf{I}_j}{\mathbf{g}_j} \right) \approx \mathbf{S}(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) \left(\frac{{}^B \hat{\mathbf{g}}_V \bar{\mathbf{D}} \mathbf{v}_j}{\mathbf{g}_j} \right) \text{ for } i = 1, 2, \dots, N_p \text{ and for } j = 1, 2, \dots, N_I.$$

$$J(\hat{\ell}, \hat{\mathbf{e}}) = \sum_{j=1}^{N_I} \left\{ \sum_{i=1}^{N_p} \mathbf{b}_{ij}^T \mathbf{b}_{ij} \right.$$

$$\left. - \sum_{i=1}^{N_p} \mathbf{b}_{ij}^T \mathbf{S}(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) \left(\sum_{i=1}^{N_p} \mathbf{S}^T(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) \mathbf{S}(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) \right)^{-1} \sum_{i=1}^{N_p} \mathbf{S}^T(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) \mathbf{b}_{ij} \right\}$$

$$\Rightarrow \mathbf{D}^V = \mathbf{G} = \underbrace{{}^B \hat{\mathbf{g}}_V \bar{\mathbf{D}} \mathbf{V}}_{\frac{6}{5} \mathbf{I}_{11}}$$

5. Minimize cost function:

$$\text{Minimize } J(\hat{\ell}, \hat{\mathbf{e}}) \Rightarrow \hat{\ell}, \hat{\mathbf{e}} \text{ (Initial guess: } \hat{\ell} = \ell_{\text{design}} \text{ and } \hat{\mathbf{e}} = \mathbf{0})$$

6. Calculate $\mathbf{g}_j = \left(\sum_{i=1}^{N_p} \mathbf{S}^T(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) \mathbf{S}(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) \right)^{-1} \left(\sum_{i=1}^{N_p} \mathbf{S}^T(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}) \mathbf{b}_{ij} \right) (6 \times 1)$ for $j = 1, 2, \dots, N_I$.

7. Define a $6 \times N_I$ matrix:

$$\mathbf{G} = [\mathbf{g}_1 \quad \mathbf{g}_2 \quad \dots \quad \mathbf{g}_{N_I}]$$

$$\mathbf{I} =$$

8. Determine ${}^B \hat{\mathbf{g}}_I$ and $\bar{\mathbf{K}}$:

$$\mathbf{F} = [\mathbf{I}_1 \quad \mathbf{I}_2 \quad \dots \quad \mathbf{I}_{N_I}]$$

$${}^B \hat{\mathbf{g}}_I \bar{\mathbf{K}} = \hat{\mathbf{H}}_I = \mathbf{G} \mathbf{F}^T (\mathbf{F} \mathbf{F}^T)^{-1}$$

$${}^B \hat{\mathbf{g}}_I = \frac{6}{5} h_{11} \text{ and } \bar{\mathbf{K}} = \frac{5}{6 \cdot h_{11}} \hat{\mathbf{H}}_I$$

9. Determine ${}^B \hat{\mathbf{g}}_V$ and $\bar{\mathbf{D}}$:

$$\mathbf{V} = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \dots \quad \mathbf{v}_{N_I}]$$

$${}^B \hat{\mathbf{g}}_V \bar{\mathbf{D}} = \hat{\mathbf{H}}_V = \mathbf{G} \mathbf{V}^T (\mathbf{V} \mathbf{V}^T)^{-1}$$

$${}^B \hat{\mathbf{g}}_V = \frac{6}{5} h_{11} \text{ and } \bar{\mathbf{D}} = \frac{5}{6 \cdot h_{11}} \hat{\mathbf{H}}_V$$